

CY 241: Nano-scale Science and Technology

Course Code	CY 241
Course Name	Nano-scale Science and Technology
L-T-P-C	3-0-0-3
Intended for	B. Tech. 2nd Year onward
Distribution	Elective
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

- **Properties**
Properties of materials with nanoscale dimensions. [4 Lectures]
- **Nanostructures**
Zero, one, two and three-dimensional nanomaterials. [2 Lectures]
- **General methods of synthesis**
General methods of synthesis of nanomaterials and nanostructures [5 Lectures].
- **Characterisation techniques**
Optical spectroscopy and microscopy, scanning probe microscopy, scanning electron microscopy, transmission electron microscopy and X-ray diffraction. [8 Lectures]
- **Inorganic nanomaterials**
Metallic nanocrystals with special emphasis on coinage metals, semiconductor nanocrystals, quantum dots, magnetic materials. [6 Lectures]
- **Carbon nanostructures**
Carbon nanotubes, graphene and fullerenes. [4 Lectures]
- **Organic and biological nanostructures** [4 Lectures]
- **Applications**
Catalysts, sensors, actuators, display systems, molecular devices and nanobiotechnology. [7 Lectures]
- Quizzes [2 hours]

Text Books

1. C. P. Poole (Jr.) and F. J. Owens, **Introduction to Nanotechnology**, Wiley Interscience, John Wiley and Sons, Hoboken, New Jersey.
2. G. A. Ozin and A. C. Arsenault, **Nanochemistry: A Chemical Approach to Nanomaterials**, RSC Publishing, Royal Society of Chemistry, U.K.
3. M.D. Ventra, S. Evoy, J.R. Heflin Jr. (Eds.), **Introduction to Nanoscale Science and Technology**, Kluwer Academic Publishers, Boston.
4. G. Cao, **Nanostructures & Nanomaterials: Synthesis, Properties & Applications**, Imperial College Press, 2004.

References

1. L. M. Liz-Marsan and P. V. Kamat, **Nanoscale Materials**, Kluwer Academic Publishers, Boston, USA.
2. D. A. Bonnell, **Scanning Probe Microscopy and Spectroscopy: Theory, Techniques and Applications**, 2nd Edition. New York, Wiley-VCH.
3. S. Amelinckx, **Electron Microscopy: Principles and Fundamentals**, Weinheim, VCH.